

Analysis of the Vapor Phase Dynamics during iCVD: Free Radical Initiation by *tert*-Butyl Peroxide

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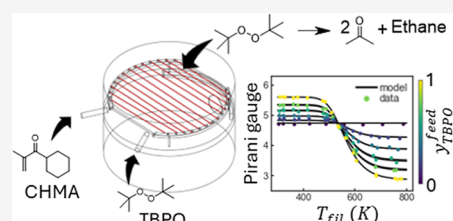


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ABSTRACT: While initiated chemical vapor deposition (iCVD) has been used in a wide range of fundamental and applied studies in the past two decades, little is known about the reactive vapor phase, leading to inconsistent treatment of the initiation process in the existing literature. Di-*tert*-butyl peroxide (TBPO) is commonly used as an initiator in iCVD by leveraging radical intermediates produced during thermal decomposition. Despite its widespread use, no models exist for TBPO decomposition in iCVD. To bridge this knowledge gap, we integrated the Pirani pressure gauge into the iCVD setup to measure the real-time vapor-phase composition in the reactor. After establishing the relationship between the Pirani gauge pressure and composition from batch data, we use it to quantify the fractional conversion of pure TBPO in the reactor in the continuous mode. We find that under iCVD conditions, TBPO fully decomposes to acetone and ethane following a first-order Arrhenius kinetics. This means that (i) the partial pressures of radicals are small (below the detection limit of the Pirani gauge) and (ii) there is a stoichiometric increase in the number of gas molecules in the reactor, which has not been accounted for in existing models. This causes the overestimation of monomer pressure, which is a key parameter in the current understanding of iCVD. We analyze the decomposition of TBPO in the presence of cyclohexyl methacrylate (CHMA) monomer suggesting that the monomer does not react appreciably in the vapor phase. To show how this model of the vapor phase is related to the deposition kinetics, we compare our data to the classical linear polymerization theory, which describes free radical polymerization of CHMA in solution and finds an acceptable fit. Notably, in Arrhenius plots of the deposition rate, there is a well-documented but unexplained decrease in the apparent activation energy at high filament temperature ($>300\text{ }^{\circ}\text{C}$), which is predicted by our models.



1. INTRODUCTION

A century ago, free radical polymerization (initiated by peroxides) was used to produce the first synthetic polymers.¹ Since then, polymers have become increasingly important in every major industry. One major application is polymer thin films, which are foundational in technologies, such as surface chemistry modification, thin-film composite membranes, organic electronic devices, and passivating coatings. The most popular polymer thin film synthesis methods are solution phase methods, such as spin coating, spin-casting, and dip coating. Unfortunately, these methods produce high-quality films only on flat and chemically compatible substrates. In response to these limitations, a diverse set of polymer chemical vapor deposition (CVD) technologies has been developed. Polymer CVD eliminates the use of solvents, produces films which conform to nanostructures, and performs well with most substrate surface chemistries. While these properties lend distinct advantages, many types of polymer CVD are restricted to specific polymer chemistries (like parylene CVD) or require harsh processing conditions that are incompatible with delicate substrates (like plasma-enhanced CVD).² Unlike other

polymer CVD methods, initiated chemical vacuum deposition (iCVD) allows independent control over the initiator and monomer chemistries. This expands the library of possible polymer chemistries while enabling mild processing conditions.^{3,4}

Since the invention of iCVD, studies investigating its kinetics have proliferated.^{5–20} These studies aim to model the iCVD process, with special focus on the roles of the initiator and monomer.³ The current understanding of initiator dependence originates primarily from studies of iCVD in nanoconfinement, where deposition in pores and trenches is described using models of the initiating radicals undergoing Knudsen diffusion.^{10,12,20} These studies support an Eley–Rideal initiation model in which the rate of initiation is independent

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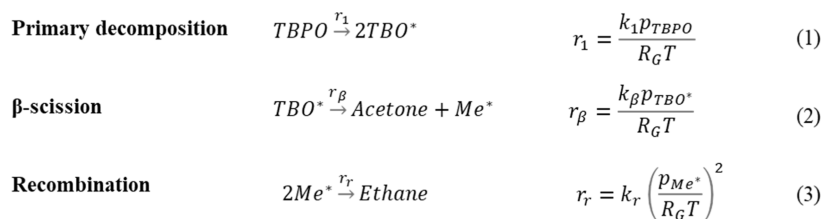
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Scheme 1. Thermal Decomposition of TBPO



of the substrate temperature. This finding suggests that changes in the deposition rate with sample temperature are due only to the monomer.

Using this idea, the role of a monomer in deposition has been examined by measuring the deposition rate over a range of stage temperatures.^{11,15,17,19,20} The results of these studies show that deposition rate follows an Arrhenius dependence on stage temperature, which matches the temperature dependence of monomer adsorption. When the monomer pressure is well below saturated conditions,²¹ the dimensionless ratio of monomer partial pressure to its saturation pressure (p_m/p_{sat}) is used to account for the role of the monomer in deposition consistent with the Brunauer–Emmett–Teller (BET) theory of adsorption.^{11,15,17,19,20} No consensus has been reached on the exact form that this dependence takes. However, the relationship between deposition rate and p_m/p_{sat} is fit as linear with either a positive^{15,20} or negative^{11,19} y-intercept, and some discussion of a quadratic dependence at low p_m/p_{sat} .^{11,19} This existing work provides strong evidence that the surface concentration of a monomer is determined by adsorption and that initiating radicals likely react upon collision with the surface (Eley–Rideal mechanism). Critically, however, this literature does not establish the order of reaction with respect to either initiator or monomer and does not comment on the potential role of vapor phase phenomena (e.g., TBPO decomposition and mixing).

So far, direct measurement of vapor phase composition during deposition is rare,²² and existing efforts to quantify it^{6,22} do not validate any specific model. While computational simulations of iCVD have been performed,^{23,24} their scope is limited, and they do not simulate typical iCVD conditions. With no experimentally validated model of the vapor phase to draw from, kinetic studies rely on largely unexamined assumptions when attempting to predict how vapor phase composition depends on flow rate or filament temperature.^{5–15} This issue impacts every study which has compared complete quantitative mechanistic models (i.e., models which relate a complete set of process variables to deposition rate) to their experimental results and has been broadly acknowledged in the existing literature.^{5,8,9} The volume of work that relies on unverified assumptions of the iCVD vapor phase illustrates a critical vulnerability and underscores the need for an experimentally validated model.

While a model of the vapor phase is necessary to further understand the mechanisms underlying the iCVD process, it is also necessary to guide the choice of initiator, understand and optimize additives like vapor solvents,²⁵ measure and engineer monomer yield,²⁶ and improve reactor control.²² In the field of inorganic CVD, extensive reactor simulations have been instrumental in guiding the optimization of reactor design parameters for next-generation reactors and enabling the direct study of surface reactions.²⁷

The key vapor-phase reaction in iCVD is thermal decomposition of the initiator. Among the various initiators used in iCVD,^{5,14} di-*tert*-butyl peroxide (TBPO) is by far the most widely studied and utilized.³ While TBPO decomposition has been characterized experimentally,^{6,22} a complete model of its decomposition has not yet been developed for the iCVD system. Outside of the iCVD literature, however, fundamental studies of reaction kinetics have extensively examined TBPO as a model gas-phase decomposition reaction.^{28–30} This work shows that TBPO decomposes via three steps (Scheme 1): (i) the primary decomposition, in which the peroxide bond is cleaved to produce *tert*-butoxy radicals (TBO*); (ii) rapid β -scission, in which TBO* radicals decompose into methyl radicals (Me*) and acetone; (iii) recombination of Me* radicals to form ethane. The primary decomposition is sufficiently slower than the β -scission and recombination reactions that the overall reaction is observed to be first order in the concentration of TBPO.²⁸

In Scheme 1, r_1 is the rate of primary decomposition in $\text{mol m}^{-3} \text{s}^{-1}$, k_1 is the rate constant for primary decomposition in s^{-1} , p_{TBPO} is the partial pressure of TBPO in Pascals, T is the temperature in Kelvin, R_G is the ideal gas constant ($8.314 \text{ J mol}^{-1} \text{ K}^{-1}$), r_β is the rate of β -scission in $\text{mol m}^{-3} \text{s}^{-1}$, k_β is the rate constant for β -scission in s^{-1} , p_{TBO^*} is the partial pressure of TBO* radicals in Pascals, r_r is the rate of recombination in $\text{mol m}^{-3} \text{s}^{-1}$, k_r is the rate constant for recombination in $\text{m}^3 \text{mol}^{-1} \text{s}^{-1}$, and p_{Me^*} is the partial pressure of Me* radicals in Pascals.

Compiled results from previous studies show the Arrhenius kinetics for the primary decomposition reaction has an activation energy, $E_{A,1}$, of $158.2 \pm 0.2 \text{ kJ mol}^{-1}$ and a pre-exponential factor, A_1 , of $10^{15.8 \pm 0.2} \text{ s}^{-1}$.²⁸ Though TBPO decomposition is an effective method of generating initiating radicals,³⁰ rapid β -scission and methyl radical recombination lead to partial pressures of the radical intermediates, which are difficult to directly measure. The partial pressures of the initiating radical species are determined by the relative rates of β -scission and recombination, so an understanding of these reactions is necessary to estimate radical availability. Because the recombination of methyl radicals is a fundamental gas phase radical reaction, it has been well characterized outside the context of TBPO thermal decomposition.^{31–33} A notable qualitative insight from the study of Me* recombination is that it proceeds more rapidly at low temperatures because of its negative reaction entropy.²⁸ Nevertheless, applying models of Me* recombination quantitatively to TBPO decomposition is uncertain because TBPO decomposition products may influence the kinetics.³¹ Unlike Me* recombination, β -scission is difficult to study outside the context of TBPO decomposition, so it is less well studied.³⁴

Characterization of the vapor phase is instrumental to developing an understanding of the decomposition of TBPO and other vapor phase processes in iCVD. Existing vapor phase

characterization methods, such as in situ FTIR⁶ and in situ mass spectrometry,²² have been used to directly characterize the vapor phase. Both are high-cost and are capable of measuring composition only in the chamber exhaust and cannot measure composition in the chamber directly, presenting barriers to broad implementation. We have introduced a method by which vapor phase composition can be deduced from the Pirani pressure gauge³⁵ measurements, which we summarize here. A Pirani gauge reports the pressure by measuring heat transfer from a fine wire. At low pressures (less than 0.5 Torr for most gases), the rate of heat transfer is limited by the rate at which gas molecules collide with the wire; therefore, it is proportional to pressure. Larger gas molecules transfer more energy per collision, so the Pirani gauge pressure will vary with composition (larger for larger gas molecules and vice versa). Leveraging this dependence, this work develops a model of how the Pirani gauge pressure depends on vapor-phase composition to enable a real-time composition measurement based on the gauge reading. Only when multiple wavelengths are detected from in situ FTIR is it qualitatively more informative than the Pirani gauge method that we present.

To move closer to a fundamental understanding of iCVD kinetics and address the absence of a well-validated vapor phase model, we used the Pirani gauge to systematically characterize the thermal decomposition of TBPO in the iCVD reactor. In agreement with the literature on gas-phase decomposition, we find that TBPO decomposes fully into acetone and ethane following Arrhenius kinetics. We further find that the iCVD reactor should be treated as a well-mixed system (frequently referred to as a continuous stirred tank model) for the purposes of TBPO decomposition (consistent with our past work).³⁵

To probe the interactions between TBPO and the monomer during iCVD, we use cyclohexyl methacrylate (CHMA) as a model system. We choose to focus on a methacrylate monomer because they are commonly used in iCVD,³ and they tend to follow classical linear polymerization kinetics^{36,37} (often considered to be “ideal”³⁸). By using a monomer with ideal kinetics, we minimize the complexity of the problem. In contrast, acrylates tend to undergo backbiting reactions, resulting in chain branching.^{39,40} Because kinetic models of acrylate polymerization must account for these backbiting reactions, the study of acrylate kinetics is often complex and difficult to generalize.⁴⁰ CHMA is an especially good choice as a model monomer because it has been studied in the iCVD literature,^{14,17} it has an inert, nonpolar side-group, it deposits at a reasonable rate (unlike methyl methacrylate,¹⁶ which is a common model system in solution phase polymerization), and it has been studied in solution and shown to follow classical linear kinetics.^{36,37,41}

We model and characterize the thermal decomposition of TBPO in the presence of CHMA vapor with a high-stage temperature, while deposition is not occurring. We modeled the Pirani gauge measurement without including vapor phase reactions between CHMA and TBPO, suggesting that they do not interact. This flow-through experiment is repeated with a low stage temperature to estimate the effect of deposition on the vapor composition. For the case we studied, neglecting deposition when calculating the mole fraction of monomer in the vapor phase would result in roughly a 5% overestimation of the monomer mole fraction.

To further explore the role of initiation in the deposition process and root our investigation in existing studies of iCVD kinetics, we fit deposition rate data to the classical linear polymerization model,³⁸ adapted for iCVD. To adapt this model from the solution phase to iCVD, we assume that surface concentrations of monomers and initiating radicals are proportional to their partial pressures. Similar assumptions are made in existing models of iCVD¹⁹ kinetics and are expected to be rigorous at low p_m/p_{sat} , such that nonlinearity from monomer adsorption is negligible. As previous work has done, the model assumes that polymerization is initiated by TBO*. This model performs reasonably against data, predicting a well-known deviation from Arrhenius kinetics for the deposition rate at a high filament temperature. We hope that our work will enable a deeper understanding of iCVD kinetics by elucidating the role of vapor phase reactions. We expect that revisiting iCVD kinetics with a deeper understanding of the vapor phase will yield fruitful insight into the fundamental processes.

2. MATERIALS AND METHODS

2.1. Materials. CHMA (99.98% purity, AmBeed) was used as a monomer, and di-*tert*-butyl peroxide (TBPO) (98%, from Sigma-Aldrich) was used as an initiator in all depositions and mixed-gas flowthrough experiments. Both were used without further purification. Depositions were performed on 0.5 mm-thick polished P/Boron-doped silicon wafers (from University Wafer).

2.2. Filament Temperature Calibration. To measure the temperature of the filament array, a k-type thermocouple was attached to the filament such that it measured temperature independently of the direction current was flowing through the array, ensuring the thermocouple was not being shorted by the array. We expect that the thermal decomposition of TBPO flowing at 1 sccm and 100 mTorr is highly repeatable, since we have good control over the decomposition conditions (flow rate, pressure, chamber temperatures, etc.); however, the measured temperature of the filament is known to be an extremely sensitive system, changing from day to day due to slight changes in thermal contact, which appear to occur even when it is welded to the wire. By using the Arrhenius relationship between filament temperature and thermal decomposition of TBPO (discussed in Section 3.2), we can perform a calibration procedure at the start of a measurement to ensure the same temperature scale is used between days.

Specifically, we have standardized all temperature measurements (unless otherwise noted) by measuring the Arrhenius curve for TBPO flowing at 1 sccm and 100 mTorr. A standard Arrhenius curve was collected (see Section 3.3 for details), which serves as a calibration data set to which all other temperature curves were calibrated. Equation 4 adjusts measured temperatures onto the consistent temperature scale from the calibration data. Equation 4 is purely empirical, and its validity as a temperature correction requires a good fit of both the calibration data and the calibrated data to the Arrhenius curve at both adjusted and unadjusted filament temperatures. All calibrated and uncalibrated data showed strong fit to the Arrhenius model. Temperature corrections of up to 100 °C were necessary for the highest temperatures but generally much less. Further details of the calibration may be found in the Supporting Information.

$$T_{\text{fil}} = \frac{E_{A,1,\text{CAL}}^{\text{eff}}}{\frac{E_{A,1}^{\text{eff}}}{T_{\text{fil,meas}}} - R_G(\ln A_1^{\text{eff}} - \ln A_{1,\text{CAL}}^{\text{eff}})} \quad (4)$$

where $E_{A,1}^{\text{eff}}$ is the apparent activation energy and A_1^{eff} is the pre-exponentiation factor. Subscript “CAL” is used to refer to the calibration data while subscript “meas” refers to the measured data. $E_{A,1,\text{CAL}}^{\text{eff}}$ and $A_{1,\text{CAL}}^{\text{eff}}$ were 91.531 kJ mol^{−1} and 4.232 × 10⁴ m³ s^{−1}, respectively.

2.3. Vapor Phase Composition Measurements Using a Pirani Gauge. A Granville-Phillips Company convection gauge (catalog no. 275) calibrated for nitrogen gas was used for composition

sensing and will be referred to throughout as the Pirani gauge. The pressure reported by it will be referred to as the Pirani pressure, while actual mechanical pressure is sensed using an MKS Baratron Capacitance Manometer model 626C, and this pressure will be referred to as the Baratron pressure.

To measure the composition using the Pirani gauge, a standard curve is created by measuring the Pirani and Baratron pressures under controlled conditions where total flow rate and reactor pressure is held constant, but makeup of the inlet gas is changed. The ratio of the Pirani pressure to the Baratron pressure is then plotted against the reactor composition (mole fraction in the case of binary gas mixing or fractional conversion in the case of a first-order reaction) and fit to a model (specific models used are discussed in the methods sections). In scenarios where the reactor composition is not known but the Pirani gauge pressure and Baratron pressures can be read, the standard curve can then be used to determine the reactor composition.

2.4. Batch Mode TBPO Decomposition. During batch decomposition (and all other experiments performed in the reactor), the temperatures of the lid, reactor body, and stage were monitored using k-type thermocouples. The reactor was charged with TBPO at a fixed pressure of 100 mTorr at which point it was held under these conditions for 1 min to allow for measurement of the leak rate. Keeping the reactor isolated from a vacuum, the filament array was turned on, and the Baratron and Pirani gauge pressures were measured as a function of time. The reactor was then pumped down to base pressure again and these procedures were repeated for decomposition at filament temperatures from 180 to 280 °C. The pressure curves are fitted to a first-order kinetic model to obtain primary decomposition rate constants for the reactor. The ratio of Pirani pressure to Baratron pressure is plotted against fractional conversion, generating a standard curve to measure fractional conversion given measurement of Pirani and Baratron pressures.

2.5. Continuous Flow-through TBPO Decomposition. Using the form of the standard curve generated from batch mode TBPO decomposition, Baratron and Pirani gauge pressures for continuous flow-through TBPO decomposition can be used to better understand the iCVD process (which occurs in flow-through mode). To test if the rate of decomposition depends on the flow rate of TBPO, the composition in the reactor was measured using the Pirani gauge at a controlled pressure of 100 mTorr and filament temperature of 240 °C while varying the flow of TBPO into the reactor over its full range from 0.11 to 1.479 sccm.

At a constant flow rate of 1 sccm and total pressure of 100 mTorr, the power to the array was then turned on so filament temperature reached ~160 °C (a low value below which the reaction is too slow to measure). Power was then increased incrementally until a filament temperature of 360 °C was reached (the highest temperature our power supply could reliably reach) while the pressures and temperatures were recorded. To ensure that sufficient time was given for thermal equilibration while stepping up the temperature, the power to the array was stepped back down while pressures and temperatures were recorded. No significant hysteresis was observed in fractional conversion; however, there was some hysteresis in measured stage temperatures. This procedure was performed at pressures of 50, 100, 200, and 300 mTorr.

A similar flow-through experiment was then performed for TBPO and CHMA mixtures at inlet TBPO mole fractions of 0, 0.2, 0.4, 0.6, 0.8, and 1 at a reactor pressure of 100 mTorr and a reactor stage temperature of 50 °C such that deposition was unlikely to impact the vapor-phase composition. Pirani gauge pressure was measured for filament temperatures from room temperature to 430 °C for each flow condition (the full range of filament temperatures accessible to our system). Time for equilibration (roughly 5 min) was allowed between flow conditions.

2.6. iCVD Polymerization. All iCVD polymerization processes were performed in a custom-built two-chamber iCVD with a volume of 6 L. Potential error in reactor volume used in calculations only has a significant effect on pre-exponential factors and is otherwise inconsequential. All samples were cut from silicon wafers to be the

same size, 50 cm², setting aside a reference sample from each wafer for baselining FTIR absorbance spectra. Samples were placed in the center of the cooled stage, and a K-type thermocouple was affixed to the stage adjacent to the sample using Kapton tape and a Cool-Grease thermal heat transfer compound from AI technologies. Once the reactor reached a base pressure of 3–6 mTorr, the leak rate was recorded by measuring the Baratron pressure every 10 s for 2 min and performing a linear regression to obtain the slope. This change in pressure over time is then converted to a molar leak rate by using the ideal gas law. For the leak rate and all flow-rate measurements, the effective temperature was assumed to be 42.5 °C as an average of the stage temperature of 30 °C and the wall/lid temperature of 50 °C (in accordance with past work on the effective temperature).³⁵ Depositions with a leak rate over 0.03 sccm were discarded from the data analysis.

Flow rates of CHMA and TBPO were determined by closing the throttling valve connecting the chamber and the vacuum pump and measuring pressure every 2 s for 20 s. The rate of change of the pressure was then used to obtain a molar flow rate using the ideal gas law.

To calibrate the Pirani gauge measurement before the deposition, Pirani pressure, filament temperature, stage temperature, voltage, and current to the array (as read by the BK precision DC power supply) were all recorded at the following set of conditions.

- Filament off, with a 1 sccm CHMA flow rate at a Baratron pressure of 100 mTorr.
- Filament off, with no CHMA flowing, with a 1 sccm TBPO flow rate at 100 mTorr.
- Filament power set to a nominal set point for the deposition, with 1 sccm TBPO at 100 mTorr.
- Filament power set to maximum with 1 sccm TBPO at 100 mTorr.

The deposition was then performed for 1 h at a pressure of 100 mTorr and a total flow rate of 1 sccm. Depositions were performed at a range of initiator/monomer inlet compositions and filament temperatures, with all other variables held constant. Deposition conditions are provided in the [Supporting Information](#).

2.7. Thin Film Characterization. A JA Woollam alpha-SE spectroscopic ellipsometer collected spectra at 65, 70, and 75° angle of incidence on poly(CHMA) thin films. Data were fitted using Complete EASE 6 software, version 6.51 with a 3-layer model of a Cauchy film on a silicon thermal oxide on a silicon metal substrate. For many samples, the Cauchy model was not appropriate due to large mean squared error (MSE), but for consistency, the same model is used throughout.

A Thermo Scientific Nicolet iS50 FT-IR collected transmission spectra of all polymer film samples and their respective reference wafers using a DGTS detector. A single grazing incidence spectrum of the CHMA monomer was taken using a variable angle grazing incidence attenuated total reflectance (ATR) module with a germanium crystal for reference and also using the DGTS detector. All spectra were collected from 700 to 4000 cm⁻¹ with a resolution of 4 cm⁻¹. Omnic software was used for atmospheric suppression, which eliminated spurious peaks due to the presence of CO₂. To determine the sample absorbance from the FTIR spectral data, quartic baseline subtraction was performed by taking the derivative of the spectrum, then fitting a third-order polynomial to the baseline and subtracting it. After removing outlier spectra, spectra were then averaged, and the average spectrum was multiplied by a coefficient and subtracted from each of the spectra individually such that the residual was minimized. The coefficient for subtraction was then reported as the “optimal scale”, which is directly proportional to the total absorbance of CHMA. No evidence of contamination was observed in sample spectra, and all residual spectra resulting from subtraction were checked to ensure minimal error. Total absorbance of CHMA was related to mass per unit area using the density of bulk poly(CHMA) and the average absorbance per nanometer of film for samples thicker than 250 nm, having MSE less than 30. A discussion of MSE can be found in the software documentation,⁴² and we have included a short

explanation of the MSE as it relates to spectroscopic ellipsometry in the [Supporting Information](#).

3. RESULTS AND DISCUSSION

3.1. Batch-Mode Decomposition of Pure TBPO. The key vapor-phase reaction in iCVD is TBPO decomposition. In a closed chamber with a fixed initial quantity of TBPO, pressure increases as TBPO decomposes because each molecule of TBPO decomposes into two molecules of acetone and one molecule of ethane. This pressure change is frequently used to characterize TBPO decomposition kinetics.²⁹ However, the conditions it is typically studied at (atmospheric pressure and uniform reactor temperature) differ significantly from iCVD.^{28,29} In iCVD, (i) the filament array introduces significant temperature nonuniformity and (ii) the leak rate contributes measurably to pressure changes. Because these factors affect the pressure, the reaction kinetics must be validated under iCVD-specific conditions. While batch experiments are illuminating, continuous flow-through experiments more accurately represent iCVD. Unfortunately, they do not allow kinetic measurements based on pressure change. Therefore, this section also establishes the relationship between the Pirani gauge reading and vapor-phase composition during TBPO decomposition, so the Pirani gauge can be used as a composition sensor.

The following steps correct for the effects of leak and nonuniform temperature, such that pressure change during batch mode decomposition experiments can be used to deduce composition in the reactor. We start by accounting for the effect of changing gas temperature on the total pressure in the reactor (i.e., effect (i) mentioned above). In past work,³⁵ we established a method to correct for temperature changes in batch mode using an effective temperature, defined such that the ideal gas law (which is used routinely in iCVD research such as precursor flow-rate calibration) holds for the reactor according to [eq 5](#)

$$T_{\text{eff}} = \frac{pV_r}{n_{\text{tot}}R_G} \quad (5)$$

where T_{eff} is the effective temperature in Kelvin, p is the total pressure in the reactor in Pascals, V_r is the reactor volume in m^3 , and n_{tot} is the total number of moles of gas in the reactor. Our previous work has measured the effective temperature using inert argon and IPA, respectively, showing that changes in gas temperature are responsible for a 5–10% change in pressure after the filament is turned on.³⁵ That work also validates a model of the effective temperature obtained by averaging T^{-1} (in Kelvin) over the volume of a 2-chamber reactor (matching our experimental setup). In this average, the bottom chamber is isolated from the filament array; the filament array is described by a planar heat source, and temperature profiles are linearized. The result is given by [eq 6](#)³⁵

$$T_{\text{eff}} = \frac{1}{\frac{1-f}{T_{\text{ref}}} + \frac{f \ln\left(1 + h_{\text{eff}}\left(\frac{T_{\text{fil}}}{T_{\text{av}}} - 1\right)\right)}{h_{\text{eff}}(T_{\text{fil}} - T_{\text{av}})}} \quad (6)$$

where T_{av} is the average temperature in Kelvin of the lid and the stage when the filament is off (assumed to be equal to the stage temperature), f is the fraction of the total reactor volume which is occupied by the filament array (estimated from the area of the filament array (371 cm^2) multiplied by the height of

the chamber and divided by the total volume of the reactor, giving 0.32), and T_{ref} is the temperature of the bottom chamber of the reactor (40°C). At especially low pressures (or Knudsen numbers), the Knudsen temperature jump^{43,44} may introduce pressure dependence into the overall heat transfer from the array. However, the data of Bakker et al.⁴⁵ and our past work³⁵ suggest that this effect is negligible at our conditions and that heat transfer between the filament and reactor body is described by an effective heat transfer coefficient (h_{eff}), given by [eq 7](#)³⁵ (details of the derivation of [eqs 6](#) and [7](#) as well as plots to aid in physical intuition may be found in the [Supporting Information](#) and in past work³⁵)

$$h_{\text{eff}} = \frac{\left(\frac{H_1 H_2}{H_1 + H_2}\right) \frac{2\pi}{L}}{\ln\left(\frac{L}{2R_{\text{fil}}\pi}\right) + \left(\frac{H_1 H_2}{H_1 + H_2}\right) \frac{2\pi}{L}} \quad (7)$$

where H_1 is the distance from the filament to the lid ($22.86 \pm 0.02 \text{ mm}$), H_2 is the distance from the filament to the stage ($31.75 \pm 0.02 \text{ mm}$), L is the mean distance between the array wires ($14.7 \pm 0.2 \text{ mm}$), and R_{fil} is the radius of the filament ($0.20 \pm 0.02 \text{ mm}$).

Batch decompositions were performed by filling the reactor with TBPO, then closing the outlet to the vacuum pump, and turning on the filament heating. As discussed previously, there is a stoichiometric pressure increase associated with decomposition of TBPO. Side reactions such as hydrogen abstraction to form *tert*-butanol are not known to occur over the pressure range considered,³⁰ so the pressure change is due only to conversion of TBPO to acetone and ethane. Correcting for (i) temperature effects and (ii) leak, the resultant pressure change can be used to experimentally determine fractional conversion (χ) using [eq 8](#)

$$\chi = \frac{1}{2y_{\text{TBPO}}^i} \left(\frac{p}{p^i} \frac{T_{\text{eff}}^i}{T_{\text{eff}}} - \frac{1}{p^i} \left(\frac{dp}{dt} \right)_{\text{leak}} t - 1 \right) \quad (8)$$

where χ is the fractional conversion of TBPO, y_{TBPO}^i is the initial mole fraction of TBPO in the reactor (generally over 0.99), p^i is the initial pressure in the reactor, $\left(\frac{dp}{dt}\right)_{\text{leak}}$ is the measured pressure change caused by leak at base pressure, with the filament off and T_{eff}^i is the initial effective temperature of the reactor (obtained from [eq 6](#) and the experimental process conditions). Details of the derivation of [eq 8](#) can be found in the [Supporting Information](#).

To model the decomposition kinetics measured by [eq 8](#), the rate equation at uniform temperature ([eq 1](#)) must be adapted for the iCVD geometry. When adapting [eq 1](#) ([Scheme 1](#)) for the iCVD geometry, we hypothesize that the total rate of primary decomposition (r_1^{tot}) is proportional to the product of the (measurable) mole fraction of TBPO far from the array and the concentration of ideal gas at the filament temperature. We introduce an effective rate constant for primary decomposition (k_1^{eff}) as a constant of proportionality that accounts for the poorly defined region close to the array. This expression is given by [eq 9](#)

$$r_1^{\text{tot}} = k_1^{\text{eff}} \frac{p}{R_G T_{\text{fil}}} y_{\text{TBPO}} \quad (9)$$

where $r_1^{\text{tot}} = \int_V r_1 dV$ in mol s^{-1} . In [eq 9](#), the effective rate constant, k_1^{eff} , is determined empirically, but represents the k_1 ,

from eq 1, averaged over the reactor. It is likely that coupled heat and mass transfer induce compositional gradients close to the filament, meaning that TBPO decomposition occurs over a range of concentrations and temperatures. Estimating k_1^{eff} from first-principles would require modeling efforts outside the scope of this paper. Instead, we simply assume that since k_1 is known to obey Arrhenius kinetics,²⁹ k_1^{eff} will as well. However, the pre-exponentiation factor and apparent activation energy will differ and are determined empirically. Because primary decomposition occurs close to the array, we use the filament temperature as the decomposition temperature, so k_1^{eff} is given by eq 10

$$k_1^{\text{eff}} = A_1^{\text{eff}} \exp\left(-\frac{E_{A,1}\eta_1}{R_G T_{\text{fil}}}\right) \quad (10)$$

where A_1^{eff} is an empirically determined pre-exponentiation factor and $E_{A,1}$ is the “true” activation energy for the decomposition of TBPO at uniform temperature (measured in the literature). To account for thermal gradients at the array, η_1 is a number between zero and one, which relates the true activation energy $E_{A,1}$ to the apparent activation energy from Arrhenius plots. A discussion on the theoretical basis for A_1^{eff} and η_1 is included in the Supporting Information. However, they are treated as empirically derived quantities in this work.

Because thermal decomposition kinetics are limited by the (unimolecular) primary decomposition of TBPO into TBO* radicals, the fractional conversion is expected to exponentially decrease according to eq 11

$$\chi = 1 - \exp\left(-\frac{k_1^{\text{eff}} T_{\text{eff}}}{T_{\text{fil}}} t\right) \quad (11)$$

where t is the time since the reaction started.

Plotting fractional conversion (calculated using eq 8) against time on linearized axes (Figure 1a) shows that when k_1^{eff} is treated as a fitting parameter, eq 11 delivers an excellent fit to the data over a range of filament temperatures ($r^2 > 0.97$ for all cases except $T_{\text{fil}} = 186.4$ °C where $r^2 = 0.95$). The success of this fit supports the assumption that TBPO decomposes directly into acetone and ethane and, therefore, that the TBO* and Me* intermediates instrumental to iCVD are only present in trace quantities. This conclusion has significant implications for kinetic modeling and reporting of deposition conditions, since we are aware of no past work which corrects for TBPO decomposition in modeling.

Plotting $\ln(k_1^{\text{eff}})$ versus $1000/T_{\text{fil}}$ shows that eq 10 is an accurate approximation of the overall decomposition rate (Figure 1b). This fit shows an apparent activation energy of 117 kJ mol⁻¹ ($\eta_1 = 0.74$) and a pre-exponential factor $\ln(A_1^{\text{eff}})$ of 23.4 (m³ s⁻¹) with $r^2 = 0.98$. Though corrected for in data analysis, the potential for thermally driven pressure changes and leak reduces the accuracy of batch measurements relative to continuous flow through measurements. Therefore, we leave a more thorough discussion of Arrhenius kinetics for our discussion of decomposition in continuous mode (Section 3.2).

We now turn our attention to the relationship between gas composition and the Pirani gauge pressure during the thermal decomposition of TBPO. In previous work, we showed that the Pirani gauge is effective for measuring the composition of simple two-vapor (binary) mixtures.³⁵ TBPO decomposition parallels this scenario. TBPO decomposition produces acetone

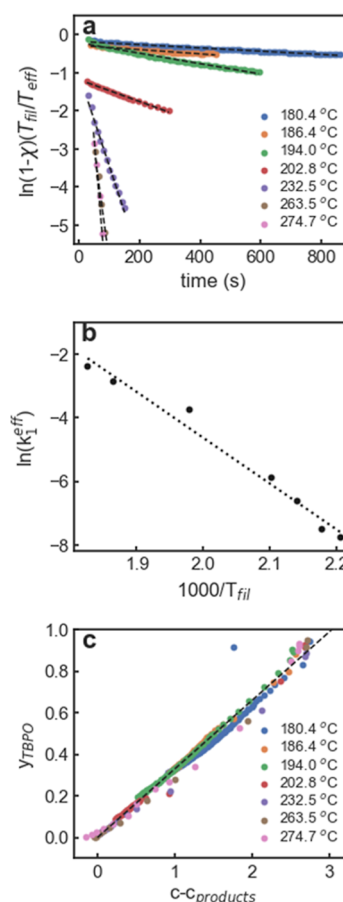


Figure 1. (a) Fractional conversion calculated from eq 8 plotted on linearized axes against time for batch-mode TBPO decomposition, for a range of filament temperatures showing a good fit to data (r^2 at least 0.95 for all cases). The fit indicates that eq 9 is appropriate to describe TBPO thermal decomposition in iCVD. (b) Arrhenius plot of k_1^{eff} from the slopes of fits in (a) in terms of filament temperature showing an apparent activation energy of 117 kJ mol⁻¹ ($\eta = 0.74$) and a pre-exponential factor $\ln(A_1^{\text{eff}})$ of 23.4 (m³ s⁻¹) with $r^2 = 0.98$. (c) Mole fraction of TBPO during the thermal decomposition of pure TBPO in batch mode calculated from eq 12 plotted against the adjusted Pirani gauge ratio $c-c_{\text{products}}$ for a range of filament temperatures including a dotted line representing eq 14 to show that eq 14 is suitable for predicting composition ($r^2 = 0.96$).

and ethane in a 2:1 ratio, so the resulting compositions in the reactor are the same as if TBPO were mixed with a 2:1 ratio of acetone and ethane. Since the Pirani gauge varies linearly with mole fraction in mixtures, we expect it to vary linearly with the mole fraction of TBPO during decomposition, which follows from the fractional conversion by eq 12.

$$y_{\text{TBPO}} = y_{\text{TBPO}}^i \frac{1 - \chi}{1 + y_{\text{TBPO}}^i 2\chi + \frac{1}{p^i} \left(\frac{dp}{dt}\right)_{\text{leak}} t} \quad (12)$$

Equation 12 is derived using a mole balance on TBPO and the definition of the fractional conversion. Details of the derivation may be found in the Supporting Information.

To correlate the composition calculated by eq 12 from the pressure change to that derived from the Pirani gauge reading, we first calculated a leak-corrected Pirani gauge ratio (the ratio of the Pirani pressure to the Baratron pressure). Leak correction is implemented in eq 13 by subtracting the rate of

pressure change caused by leak for the Pirani and Baratron gauges from their respective measured pressures

$$c = \frac{p_{\text{Pirani}} - \left(\frac{dp_{\text{Pirani}}}{dt} \right)_{\text{leak}} \frac{T_{\text{eff}}}{T_{\text{eff}}^i} t}{p - \left(\frac{dp}{dt} \right)_{\text{leak}} \frac{T_{\text{eff}}}{T_{\text{eff}}^i} t} \quad (13)$$

where $\left(\frac{dp_{\text{Pirani}}}{dt} \right)_{\text{leak}}$ is the pressure change during leak-rate

measurement for the Pirani gauge, c is the leak-corrected Pirani gauge ratio, and p_{Pirani} is the pressure measured by the Pirani gauge. A short discussion of eq 13 may be found in the [Supporting Information](#). Following the analogy to binary mixing, we propose eq 14 to relate the Pirani gauge ratio, c , to the mole fraction of TBPO in the reactor when pure TBPO is decomposing

$$y_{\text{TBPO}} = \frac{c - c_{\text{products}}}{c_{\text{TBPO}} - c_{\text{products}}} \quad (14)$$

where products subscript refers to the degradation products of TBPO decomposition (2:1 acetone to ethane).

Figure 1b shows that the composition predicted by eqs 8 and 12, based on pressure change, exhibits a linear relationship with the Pirani gauge ratio ($r^2 = 0.96$), consistent with eq 14. On different days, the Pirani gauge reports slightly different pressures for the same vapor at identical conditions. This drift is likely caused by small quantities of polymeric material, which have an outsized impact on the gauge reading. Drift is corrected in Figure 1b by subtracting the Pirani gauge ratio of the fully decomposed TBPO (c_{products}) so that the data fall on a line. In Figure 1b, c_{TBPO} ranged from 5.27 to 5.67, while c_{products} ranged from 2.24 to 2.63. Figure 1b suggests two primary conclusions. First, the Pirani gauge can be used to quantify the fractional conversion of TBPO during thermal decomposition. Second, the relationship between the Pirani gauge pressure and composition is given by eq 14. A notable curvature in some data sets is explained by contamination by leak and latency in the filament temperature readings. To ensure accurate results, the Pirani gauge is recalibrated before performing experiments, so c_{TBPO} and c_{products} are determined from the Pirani gauge pressure of pure TBPO before performing experiments. The ability to measure TBPO fractional conversion with the Pirani gauge suggested by Figure 1b enables us to study the thermal decomposition of TBPO in steady-state continuous operation of the reactor, which is covered in Section 3.3.

3.2. Flow-through Decomposition of Pure TBPO. The strong correlation provided by eq 14 in Figure 1c validates the use of a linear model relating the mole fraction of unreacted TBPO to the Pirani gauge ratio. This section capitalizes on this relationship to extend the study of TBPO decomposition to continuous flow-through mode. The continuous mode corresponds more closely to normal iCVD operation and provides explicit insight into the role that flow rate and pressure play in TBPO decomposition during iCVD.

Though the Pirani gauge provides a reliable composition measurement during continuous flow-through operation of the reactor, we are ultimately interested in evaluating the kinetic model given by eq 9. To relate the reactor composition to the kinetic model, the total rate of reaction needs to be determined from a mole balance on the reactor. Because this mole balance requires the composition of the reactor outlet, the degree of mixing in the reactor must be known.⁴⁶ For the iCVD vapor

phase, our past work has shown that the degree of mixing is determined by the Peclet number, which is expected to range from 10^{-6} to 10, such that the reactor can generally be approximated as well-mixed.³⁵ Throughout this work, we will make the well-mixed assumption (for all species except radical species, which are present in low concentrations) such that composition is constant throughout the volume of the reactor. This assumption enables prediction of k_1^{eff} according to eq 15 (derivation included in the [Supporting Information](#))

$$k_1^{\text{eff}} = \frac{F_{\text{TBPO}}^{\text{in}} R_G T_{\text{fil}} (1 - y_{\text{TBPO}})}{p y_{\text{TBPO}} (2y_{\text{TBPO}} + 1)} \quad (15)$$

where $F_{\text{TBPO}}^{\text{in}}$ is the flow rate of TBPO into the reactor, y_{TBPO} is the mole fraction of TBPO in the reactor, calculated from c (the Pirani gauge ratio) using eq 14. To determine y_{TBPO} from c using eq 14 requires c_{TBPO} and c_{products} . The value for c_{TBPO} can be measured directly from unreacted TBPO; however, determining c_{products} accurately is less straightforward. When the filament is at its maximum temperature ($\sim 400^\circ\text{C}$), almost all TBPO in the reactor has been converted to acetone and ethane, so the gauge ratio at these conditions is a close approximation of c_{products} . However, because c_{products} cannot be measured exactly, it is treated as a free parameter in future data fitting. The fitted values of c_{products} were all found to differ by only about 0.05 from the measured value of c at the maximum filament temperature.

With a way to calculate k_1^{eff} from c , we must now determine how it depends on temperature, pressure, and flow rate. In Section 3.1, eq 10 was shown to accurately represent the temperature dependence of k_1^{eff} , however, the dependence of A_1^{eff} and η_1 on pressure and flow rate was not investigated. Experimentally, we leveraged the Pirani gauge ratio to determine the dependence of thermal decomposition on the filament temperature and chamber pressure. The measurements were performed with a TBPO flow rate of 1 sccm, spanning filament temperatures from 40 to 360°C and chamber pressures from 50 to 300 mTorr. Using eqs 15 and (14), we plot $\ln(k_1^{\text{eff}})$ against $1000/T_{\text{fil}}$, showing clear Arrhenius kinetics and no pressure dependence (Figure 2a). Pressure independence is notable as it indicates that for pressures above 50 mTorr, pressure-dependent heat transfer, which is often observed in rarefied gases, can be neglected.^{35,44,45} This supports a simplified view of heat transfer and TBPO thermal decomposition for iCVD under most deposition conditions and lends credence to the use of eq 6 for the effective temperature in Section 3.1. Pressure is held constant in all other flow-through experiments, so the pressure dependence is not considered outside of Figure 2a.

When we turn our attention to the temperature dependence of TBPO decomposition, an issue arises. As is standard in the iCVD literature, we have measured the temperature of the filament array using a thermocouple. However, because the energy balance on the filament is extremely sensitive and a thermocouple disrupts that energy balance in slightly different ways during different experiments, temperatures reported across experiments are different (though they are consistent within experiments). The mechanisms by which error is introduced primarily lower the measured temperature; therefore, simply averaging many observations will not improve accuracy. Instead, the same TBPO decomposition is performed before every flow-through experiment (the 100 mTorr condition in Figure 2a), assuming any differences in the

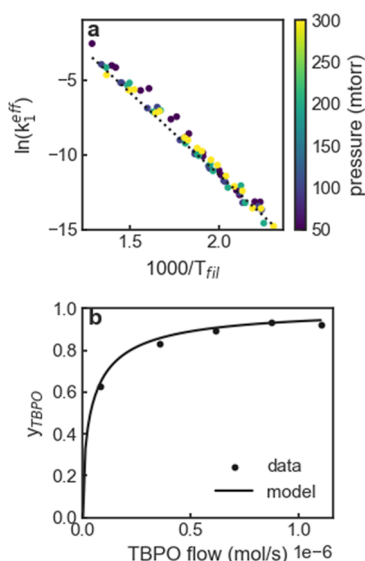


Figure 2. (a) Arrhenius plot for thermal decomposition of TBPO in a flow-through model showing pressure independence. (b) Mole fraction of TBPO from eq 14 versus flow of TBPO into the reactor including a fit from eqs 10 and (15) (solid line) where A' and η were determined from the fit performed in (a) showing the CSTR model describes flow-rate dependence of TBPO decomposition kinetics.

Arrhenius parameters are caused by temperature bias, the temperature for all-experiments is recalibrated to a consistent temperature scale using eq 4 (methods described in Section 2.2). The conditions with the highest filament temperature (assumed to be least biased) were used for calibration, having an apparent activation energy of 91.5 kJ mol^{-1} ($\eta_1 = 0.578$) and a pre-exponential factor $\ln(A_1^{\text{eff}})$ of $10.7 \text{ (m}^3 \text{ s}^{-1})$. The activation energy reported in literature for the thermal decomposition of TBPO at uniform temperature is $158.2 \text{ kJ mol}^{-1}$.²⁸ The depressed activation energy is primarily due to averaging over a range of temperatures, which is implicit in eqs 9 and 10. We did not recalibrate batch experiments since the calibration experiment was performed in the continuous mode, therefore the larger activation energy and larger pre-exponential factor in batch mode is likely caused by bias in temperature measurement. Having addressed the issue of measurement error, we can now move onto discussion of pressure and flow-rate dependence.

Pressure dependence of Arrhenius kinetics was investigated by comparing $\ln(k_1^{\text{eff}})$ versus $1000/T_{\text{fil}}$ for pure TBPO flowing at 1 sccm and 50, 100, 200, and 300 mTorr in Figure 2a, showing they correspond closely. Comparing the Arrhenius parameters for the 100 mTorr case ($\eta_1 = 0.578$, $\ln(A_1^{\text{eff}}) = 10.7 \text{ (m}^3 \text{ s}^{-1})$), the aggregate data at all four pressures give an r^2 value of 0.98, showing an excellent fit.

While the fit to Arrhenius kinetics is a strong indication that the well-mixed assumption is valid for the reactor (because of its use of eq 15), the data were all collected at a single flow rate of 1 sccm, so verification over a range of flow rates is also warranted. To do this, the Pirani gauge pressure was measured for TBPO flow rates from 0.1 to 1.5 sccm (the maximum range our system is capable of) and used to determine the mole fraction of TBPO. The rate constant for the primary decomposition of TBPO was then calculated using eq 15. Figure 2b shows the mole fraction of TBPO plotted against the flow rate with model prediction from eqs 10 and 15, where A_1^{eff} and η_1 were taken from the fit in Figure 2a. This fit

validates the use of the well-mixed assumption over the flow rates of interest. To quantify the effect of poor mixing and determine the onset of nonideality, similar experiments should be performed at higher flow rates (see our prior work for details on the effects of flow rates on mixing³⁵).

3.3. Relationship between Pirani Gauge and Composition. Batch TBPO decomposition data (Section 3.1) show that a simple linear model relates the Pirani gauge ratio to the mole fraction of TBPO during decomposition. Additionally, we examined the dependence of TBPO decomposition on filament temperature, pressure, and flow rate under continuous flow-through conditions, mimicking deposition conditions. To fully understand the vapor phase during iCVD however, it is necessary to understand thermal decomposition of TBPO in the presence of a monomer. Because CHMA is well studied in iCVD^{14,17} and is known to follow classical linear polymerization kinetics in solution,^{36,37,41} we have used it as a model monomer. This section extends eq 14 to incorporate the presence of a monomer during TBPO decomposition so that the Pirani gauge can be used to determine vapor-phase composition when CHMA is present in addition to TBPO. In the model, we hypothesize that CHMA does not react significantly with TBPO or any of its degradation products in the vapor phase. Section 3.4 evaluates this hypothesis and the model developed in this section against experimental data.

Mixing of CHMA and TBPO is directly measurable using methods discussed in our previous work³⁵ but warrants additional background on the operating principle of the Pirani gauge. The Pirani gauge measures heat loss from a fine wire and is useful as a pressure sensor at low pressures where heat transfer is proportional to pressure.⁴⁷ At these conditions, the Pirani gauge ratio for pure vapors does not vary with pressure and is expected to vary linearly with mole fraction in binary mixtures.⁴⁸ For relatively small, weakly interacting molecules (IPA and argon), we have directly observed this linear relationship.³⁵ However, when we plot the Pirani gauge ratio without thermal decomposition (c^0) against the inlet mole fraction determined by flow-rate calibration for CHMA and TBPO at 80, 100, and 120 mTorr (Figure 3a), the relationship is distinctly nonlinear for the 100 and 120 mTorr conditions. These results were repeatable on separate days. The clear pressure dependence of the Pirani gauge ratio for pure CHMA (Figure 3a) suggests that the nonlinear relationship between composition and the Pirani gauge ratio is because the Pirani gauge is operating outside its ideal range.

We have found empirically that eq 16 describes the nonlinearity observed in Figure 3a and has been included as a curve fit to data for all three pressures

$$\frac{c^0 - c_{\text{CHMA}}}{c_{\text{TBPO}} - c_{\text{CHMA}}} = \frac{y_{\text{TBPO}}^{\text{in}}}{y_{\text{TBPO}}^{\text{in}} + D^0(1 - y_{\text{TBPO}}^{\text{in}})} \quad (16)$$

where D is a fitting coefficient and superscript “0” indicates the quantity refers to mixtures of unreacted vapors, c_{CHMA} is the Pirani gauge ratio of pure CHMA, c_{TBPO} is the Pirani gauge ratio of pure TBPO, and $y_{\text{TBPO}}^{\text{in}}$ is the mole fraction of TBPO at the inlet of the reactor, calculated from flow rate calibration. A fit of eq 16 to data at 100 mTorr is shown in Figure 3a in orange and Figure 3b in red with $D^0 = 1.53$. Figure 3b also labels the points corresponding to c_{CHMA} c_{TBPO} for clarity.

Analogous to eq 16, a similar mixing curve applies to the reaction products of TBPO (2:1 acetone to ethane mixture)

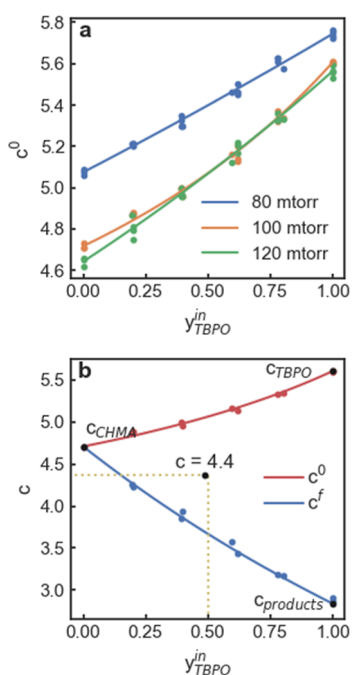


Figure 3. (a) Pirani gauge ratio in the absence of thermal decomposition (c^0) versus mole fraction of TBPO for a range of total reactor pressures from 80 to 120 mTorr showing that the Pirani gauge ratio for pure CHMA varies significantly with pressure and that there is a nonlinear relationship between Pirani gauge ratio and composition. Solid lines are fits from eq 16. (b) Pirani gauge ratio versus inlet TBPO mole fraction for unreacted TBPO with a fit of eq 16 in red and fully reacted TBPO with a fit of eq 17 in blue (superscript “ f ” indicates the quantity applies to the mixing of CHMA with the reaction products of TBPO). Pirani gauge ratio for pure CHMA (c_{CHMA}), pure TBPO (c_{TBPO}), degradation products of TBPO ($c_{products}$), and a test point to show how eq 19 is used are also included in black and labeled.

with CHMA. Correcting for the stoichiometry of TBPO decomposition, we obtain eq 17

$$\frac{c^f - c_{CHMA}}{c_{products} - c_{CHMA}} = \frac{y_{TBPO}^{in,eff}}{y_{TBPO}^{in,eff} + \frac{D^f}{3}(1 - y_{TBPO}^{in,eff})} \quad (17)$$

where superscript “ f ” indicates that the quantity applies to the mixing of CHMA with the reaction products of TBPO and $y_{TBPO}^{in,eff}$ is the effective inlet mole fraction of TBPO. When there is no deposition, $y_{TBPO}^{in} = y_{TBPO}^{in,eff}$; however $y_{TBPO}^{in,eff}$ needs to be adjusted when deposition is occurring. In the reactor mass balance, loss of monomer to deposition is equivalent to reducing the inlet monomer flow rate, so the same mixing model can be used (eq 17), provided that $y_{TBPO}^{in,eff}$ is adjusted for the loss of monomer to deposition using eq 18

$$y_{TBPO}^{in,eff} = \frac{y_{TBPO}^{in} F_{tot}^{in}}{F_{tot}^{in} - r_p^{tot}} \quad (18)$$

where r_p^{tot} is the overall rate of polymerization in the reactor in mol s^{-1} . Use of eq 18 fully accounts for deposition in the mixing model. A fit of eq 17 to data at 100 mTorr and a stage temperature of 50 °C (r_p^{tot} is zero) is shown in Figure 3b in blue with $D^f = 2.40$. Also note that the point corresponding to $c_{products}$ is labeled in Figure 3b for the sake of clarity.

Equations 16–18 relate the Pirani gauge ratio to composition for unreacted and fully reacted gases but do not allow interpolation to intermediately decomposed states. Recall from Section 3.2 that the linear model of neat TBPO decomposition (eq 14) was justified by considering that the evolution of the vapor-phase composition during the decomposition of TBPO is analogous to a binary mixing of TBPO and a 2:1 mixture of acetone to ethane. With the addition of monomer to the reactor, the change in c as TBPO decomposes is now influenced by the mole fraction of the monomer in the reactor. A detailed model of these relationships requires further work; however, as an approximation, we assume the relationship between the Pirani gauge ratio and the mole fraction of TBPO is approximately linear, such that it is described by eq 19.

$$y_{TBPO} = \frac{c - c^f}{c^0 - c^f} y_{TBPO}^{in,eff} \quad (19)$$

When pure TBPO is decomposing, $y_{TBPO}^{in,eff}$ is one, and eq 19 is identical to eq 14.

When taken together, eqs 16–19 provide a model, which relates the value of the Pirani gauge ratio to the composition in the reactor for any mixture of CHMA and TBPO at any (accessible) filament temperature, provided c_{TBPO} , $c_{products}$, c_{CHMA} , D^0 , and D^f are known at the pressure that data are being taken at (100 mTorr in our case from Figures 3a and 1c). The value of r_p^{tot} under each condition must be determined independently. We have not combined eqs 16–19 since the result is cumbersome and does not provide additional insights. However, Figure 3b is included to show how these equations relate “ c ” and y_{TBPO}^{in} to y_{TBPO} . The red line in Figure 3b represents c^0 and the blue line represents c^f . Consider the situation in which y_{TBPO}^{in} is known to be 0.5, c is measured to be 4.4 and there is no deposition occurring (shown by the labeled point in Figure 3b). First, since $r_p^{tot} = 0$, $y_{TBPO}^{in,eff} = y_{TBPO}^{in}$. To find y_{TBPO} , we see that the point of interest is midway between the red and blue lines, so we can use eq 19 to find $\frac{y_{TBPO}}{y_{TBPO}^{in,eff}} = 0.5$. Plugging in $y_{TBPO}^{in,eff}$, we find y_{TBPO} is 0.25 for this case.

3.4. Flow-through Decomposition of CHMA/TBPO Mixtures. The investigation of TBPO decomposition in the continuous flow-through mode in Section 3.2 validates the use of Arrhenius kinetics and the well-mixed assumption for TBPO and its decomposition products. However, these experiments do not provide insight into the potential interactions between TBPO and CHMA occurring in the vapor phase. Understanding how TBPO and CHMA interact is paramount, as TBPO is always accompanied by monomer vapor during iCVD. Overall, we aim to test the null hypothesis that CHMA is present as an inert gas (neither reacts with TBPO, nor influences its Arrhenius kinetics) by comparing predicted values of the Pirani gauge ratio to measured values. The model developed in Section 3.3 extends the relationship between the Pirani gauge ratio and mole fraction of TBPO during decomposition of pure TBPO (eq 14) to TBPO decomposition in the presence of CHMA (eqs 16–19). This section extends the kinetic relationship for neat TBPO decomposition (eq 15) to TBPO decomposition in the presence of CHMA. Doing so means that the Pirani gauge ratio during TBPO decomposition in the presence of CHMA can be predicted by combining the relationship between the Pirani gauge and

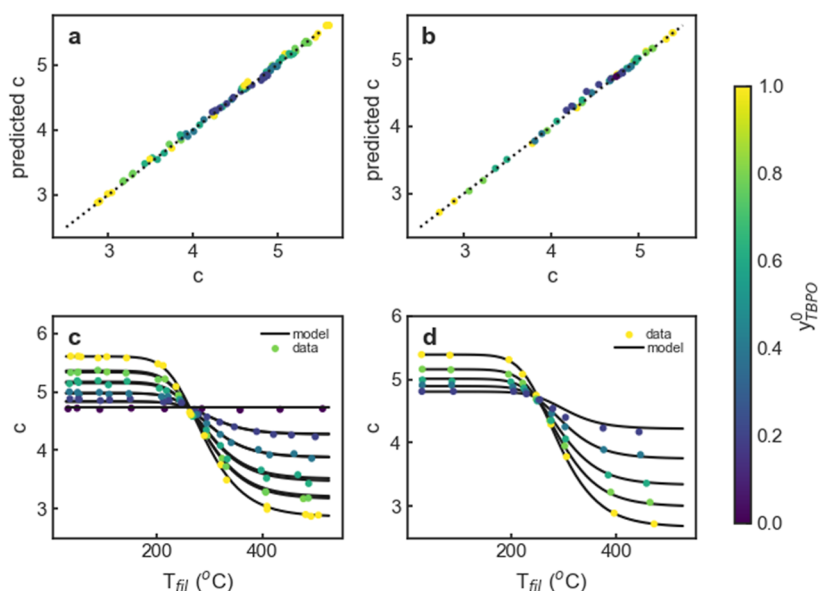


Figure 4. (a) Predicted Pirani gauge ratio (using eqs 16–20) versus actual Pirani gauge ratio for thermal decomposition of TBPO in the presence of CHMA at stage temperature of 50 °C (b) Predicted Pirani gauge ratio (using eqs 16–20) versus actual Pirani gauge ratio for thermal decomposition of TBPO in the presence of CHMA at stage temperature of 30 °C, showing a slight deviation for low inlet TBPO mole fraction (c) Measured Pirani gauge ratio (color dots) versus filament temperature for data taken at a stage temperature of 50 °C; model prediction (solid lines) was obtained using eqs 16–20, which assumes no gas phase reactions other than TBPO decomposition and that deposition does not affect vapor. (d) Measured Pirani gauge ratio (color dots) versus filament temperature for data taken at a stage temperature of 30 °C; model prediction (solid lines) was obtained using eqs 16–20, which assumes no gas phase reactions other than TBPO decomposition and that deposition does not affect vapor.

composition (eqs 16–19), and the kinetic model (discussed below). This prediction allows us to test whether CHMA can be described as an inert gas and provides a useful model for predicting vapor composition.

Extending the kinetic model for neat TBPO to mixed TBPO and CHMA is relatively simple, only requiring additional terms for CHMA flow into and out of the reactor and consumption of the monomer by deposition. Combining the rate equations given in Scheme 1 and the vapor-phase mole balance on the reactor yields eq 20 for the mole fraction of TBPO

$$y_{\text{TBPO}} = \frac{-\left(F_{\text{in}}^{\text{tot}} - r_{\text{p}}^{\text{tot}} + \frac{k_{\text{1}}^{\text{eff}} p}{R_{\text{G}} T_{\text{fil}}}\right) + \sqrt{\left(F_{\text{in}}^{\text{tot}} - r_{\text{p}}^{\text{tot}} + \frac{k_{\text{1}}^{\text{eff}} p}{R_{\text{G}} T_{\text{fil}}}\right)^2 + F_{\text{in}}^{\text{tot}} y_{\text{TBPO}}^0 8 \frac{k_{\text{1}}^{\text{eff}} p}{R_{\text{G}} T_{\text{fil}}}}}{4 \frac{k_{\text{1}}^{\text{eff}} p}{R_{\text{G}} T_{\text{fil}}}} \quad (20)$$

A detailed derivation of eq 20 may be found in the Supporting Information.

We tested the validity of eqs 16–20 (and by extension the hypothesis that CHMA is inert) by considering the decomposition of TBPO/CHMA mixtures at a stage temperature of 50 °C such that negligible deposition occurs (i.e., there is no visible buildup of polymer on any surfaces in the reactor). Equation 20 can then be used to predict y_{TBPO} based only on the inlet mole fraction of TBPO and the rate constant for primary decomposition determined from Arrhenius kinetics for pure TBPO (Section 3.2). Equations 16–19 from Section 3.3 predict the Pirani gauge ratio from the TBPO mole fraction, enabling a comparison with data. To validate eq 20, we measured Pirani gauge ratios for the full range of inlet TBPO mole fractions (0 to 1), and the full range of filament temperatures accessible to our system (25–450 °C). Together with the flow rate and pressure dependence established in

Section 3.2, this section covers the common vapor phase conditions in iCVD.

Figure 4a plots the expected Pirani gauge ratio against the measured gauge ratio, showing excellent correspondence. While mass spectrometry or FTIR is needed for further confirmation, there is no evidence in our data suggesting appreciable interaction between CHMA and TBPO. Note that the prediction in Figure 4a relies on parameters derived from three earlier fits, one in Section 3.2 to determine Arrhenius kinetics of pure TBPO decomposition and the other two from fits of eqs 16 and 17 discussed in Section 3.3 to obtain D^0 and D^f . The fit for D^0 was performed by using a gradient descent method to minimize the MSE between the data and prediction in Figure 3a, while the fit for D^f was performed by assuming that TBPO was fully decomposed at the highest achievable temperature and performing an analogous fit of eq 17 to the Pirani gauge data for the fully reacted state. The values of c_{CHMA} , c_{TBPO} , c_{products} , D^0 , and D^f used in Figure 4a were 4.71, 5.61, 2.86, 1.53, and 2.40, respectively. The relationships between these quantities can be seen clearly in Figure 3b.

While prior literature has theorized that the rate of deposition is much slower than the flow of monomer into the reactor,⁴⁹ the Pirani gauge enables a direct test of this by observing deviation from eq 20 when the Pirani gauge ratio is measured under conditions which favor deposition. When we reduce the stage temperature to 30 °C (Figure 4b), we see slight disagreement between the predicted gauge ratio and the measured gauge ratio at an inlet monomer fraction of 0.8. The values of c_{CHMA} , c_{TBPO} , c_{products} , D^0 , and D^f used in Figure 4b were 4.74, 5.39, 2.67, 2.19, and 2.19, respectively.

To help visualize the data in Figure 4a,b, we have also plotted the Pirani gauge ratio versus filament temperature for both cases in Figure 4c,d. Figure 4c shows excellent agreement over all conditions between the model and data. As expected,

Figure 4d shows slight overprediction of the Pirani gauge ratio under conditions of high monomer partial pressure and high filament temperature, which are the conditions which favor deposition.

Disagreement between data and prediction in Figure 4b,d illustrates that deposition likely impacts the vapor phase composition, but it is not immediately clear whether the effect on the vapor phase is significant. Furthermore, Figure 4 does not show how the vapor phase model given by eq 20 compares with existing methods. Because the magnitude of the impacts of deposition and TBPO decomposition on vapor phase compositions directly motivates the use of vapor phase models, further analysis is warranted. For clarity, and to minimize the impact of measurement error, we will focus on conditions which most favored deposition, 30 °C stage temperature (Figure 4b,d) and inlet TBPO mole fraction of 0.2. Because of the conclusive importance of the monomer partial pressure in iCVD literature,³ we will compare monomer mole fraction predicted using 3 methods to demonstrate the relative impact of vapor phase processes. The current mainstay method of predicting vapor phase composition (monomer composition) in iCVD assumes the composition is equal to that of the reactor feed regardless of filament temperature.³ This method neglects the stoichiometry of TBPO decomposition and the impact of deposition. The mole fraction of CHMA predicted using the standard method is represented by the dotted line in Figure 5.

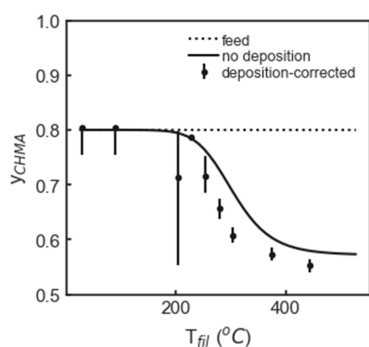


Figure 5. Plot showing the mole fraction of CHMA predicted by the feed composition alone, accounting for stoichiometry of TBPO decomposition, but not deposition, and accounting for deposition quantified by the experimentally obtained deviations from the prediction in Figure 4b,d.

To predict the CHMA mole fraction, we perform a species balance on CHMA, making the same assumptions as eq 20 to obtain eq 21.

$$y_{\text{CHMA}} = \frac{(1 - y_{\text{TBPO}}^{\text{in}})F_{\text{in}}^{\text{tot}} - r_{\text{p}}^{\text{tot}}}{F_{\text{in}}^{\text{tot}} - r_{\text{p}}^{\text{tot}} + 2 \frac{p_{\text{C}}^{\text{eff}}}{R_{\text{G}} T_{\text{fil}}} y_{\text{TBPO}}} \quad (21)$$

Details of the derivations of eqs 20 and 21 have been included in the Supporting Information. Correcting for TBPO decomposition using eqs 20 and 21, with $r_{\text{p}}^{\text{tot}}$ equal zero yields the solid line in Figure 5, showing significantly lower monomer mole fraction (by up to 0.2) compared with the standard method.

To estimate the impact of deposition on the vapor phase composition, we first predict the deposition rate necessary to cause the deviation between the measured Pirani gauge ratio

and the model prediction seen in Figure 4b,d. To accomplish this, a gradient descent algorithm varied $r_{\text{p}}^{\text{tot}}$ to minimize the difference between y_{TBPO} calculated from the Pirani gauge reading (using eqs 16–19) and y_{TBPO} calculated from eq 20 (derived using TBPO decomposition kinetics and species balance) until a difference of less than 10^{-5} was achieved. The values of $r_{\text{p}}^{\text{tot}}$ obtained represented at most 25% of the inlet monomer flow rate or 0.2 sccm of monomer lost to deposition (for the 0.8 sccm CHMA flow). To illustrate the degree to which deposition removes the monomer from the vapor phase, we calculate the mole fraction of monomer in the vapor from eq 21 for an inlet TBPO mole fraction of 0.2. The deposition-corrected CHMA mole fraction from the Pirani gauge data obtained at low-stage temperature (30 °C) is shown by data points in Figure 5, highlighting that the potential effects of deposition on the mole fraction of monomer in the reactor may be significant, further reducing the mole fraction by 0.05 in this case.

Error bars are based on measurement uncertainty for the Pirani gauge determined from past work (e.g., spread of data points in Figure 3), giving error in c of ± 0.02 . Note that the large error for data taken at a filament temperature of 200 °C (Figure 5) occurs because the Pirani gauge reading is relatively composition-independent and will have the same value for a range of compositions (Figure 4c,d shows this clearly). This illustrates the limitation of the Pirani gauge for handling uncertainty in inlet composition at this filament temperature. If the feed composition is known with good confidence, however, the Pirani gauge remains an effective metric for quantifying vapor phase composition using eqs 16–19.

Figure 5 enables two notable insights. First, the standard method of approximating the partial pressure of monomer from the feed composition leads to significant errors at conditions relevant to iCVD because it neglects the stoichiometric expansion of decomposing TBPO. Second, the effect of deposition (estimated using the method discussed above) is much smaller but still non-negligible.

3.5. Deposition Kinetics. One of the primary motivations in developing the model presented in Sections 3.1–3.4, is to help improve our understanding of the polymerization process occurring in iCVD. Therefore, this section further expands the model of the vapor phase to include radical species and develops a simple surface polymerization model that can be compared with measured deposition rates.

Because the characterization of deposited polymer requires the use of FTIR and ellipsometry, some discussion of these methods is needed. Figure 6 shows the FTIR of CHMA monomer and polymer, showing the disappearance of the vinyl bond peak at around 1700 cm^{-1} , indicating that polymerization has occurred. The FTIR spectrum for the CHMA monomer was obtained using ATR, while the poly(CHMA) spectrum was collected in transmission mode. ATR causes distortion in relative peak areas,⁵⁰ which likely causes the apparent increase in intensity of C–H peaks at 3000 cm^{-1} between monomer and polymer spectra. A more detailed FTIR peak assignment and composition analysis for poly(CHMA) can be found in Xu and Gleason (2011).¹⁴

Deposition rate data were determined for CHMA at a range of inlet mole fractions ($y_{\text{TBPO}}^{\text{in}}$) from 0.1 to 0.8 and filament temperatures from 200 to 350 °C. Deposition conditions including $p_{\text{m}}/p_{\text{sat}}$ for each condition may be found in the Supporting Information. Above an inlet TBPO mole fraction of 0.6, deposition was slow, and many samples showed a foggy

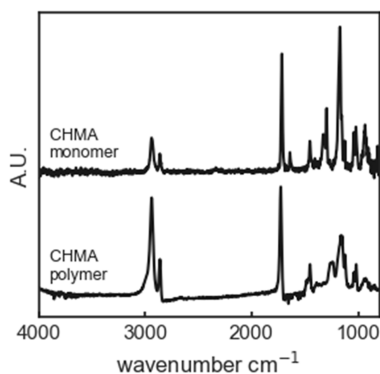


Figure 6. FTIR absorbance spectra for the CHMA monomer and CHMA polymer showing the disappearance of the vinyl peak at 1700 cm^{-1} , consistent with complete polymerization.

appearance, which may be caused by low-molecular-weight poly(CHMA). The inlet composition used in calculations was determined from flow calibration; however, the Pirani gauge ratio could also be used to determine composition (see Figure 3a). Figure 7a plots the composition derived from the Pirani gauge ratio against the TBPO inlet composition for all depositions. All points were taken at 100 mTorr and at a stage temperature of 30 $^{\circ}\text{C}$, so they have been compared to the model prediction of eq 16, using a value of D^0 (2.19) corresponding to 30 $^{\circ}\text{C}$ stage temperature experiments (Figure

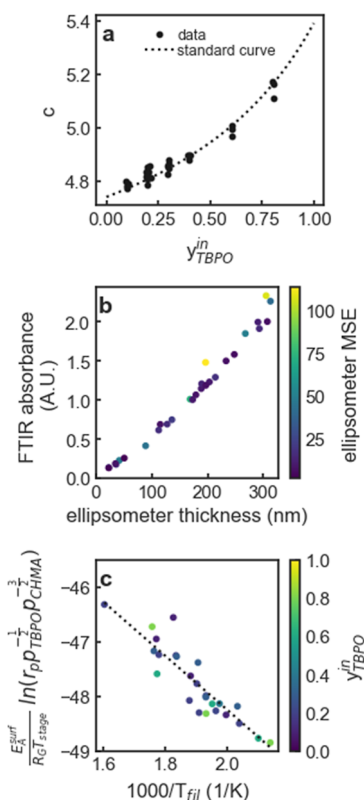


Figure 7. (a) Pirani gauge ratio versus inlet mole fraction of TBPO aligns well with the predicted value from Figure 3a (100 mTorr case). Deviation from the model may be caused by insufficient equilibration time before measurement. (b) Dimensionless FTIR absorbance versus thickness measured by an ellipsometer showing a clear nonlinear relationship, indicative of density variation with thickness. (c) Arrhenius plot for deposition rate constant based on eqs 27 and 28.

4b,d). The values of c_{CHMA} and c_{TBPO} were measured on the same day and are reported along with other deposition conditions in the Supporting Information. While the fit in Figure 7a is reasonably good, deviation between the model and the data may be due to an insufficient equilibration time.

Since the deposition rate is obtained by dividing the total quantity of accumulated polymer (per unit area) by the deposition time, an accurate measurement of film mass is needed. If film density is constant and equal to the bulk polymer density, then film thickness (from ellipsometry) can be used to estimate deposition rate; however this method is subject to error since polymer thin film density often varies with film thickness,⁵¹ and film nonideality can make ellipsometric measurements unreliable. To avoid these issues, FTIR absorbance, which is directly proportional to the area density of polymer subunits (polymer mass) by the Beer–Lambert law, was used to determine deposition rate instead. Section 2.7 details the procedures for obtaining FTIR absorbance (in arbitrary units). Figure 7b shows the plot of thickness measured by ellipsometry versus FTIR absorbance, showing apparent curvature for films less than 200 nm. This indicates that density increases with film thickness, a well-known phenomenon caused by surface excesses,⁵² though variability in molecular weight may play a small role as well. Deposition rate in $\text{mol s}^{-1} \text{m}^{-2}$ was calculated from FTIR absorption using the mean absorption per nm of thickness for films thicker than 250 nm with ellipsometer MSE less than 30, molecular weight of CHMA monomer (168.2 g mol^{-1}) and the density of bulk poly(CHMA), taken as 1.1 g mL^{-1} based on supplier specifications for commercially available poly(CHMA) to give a conversion factor of $9.84 \times 10^{-8} \text{ mol absorbance}^{-1} \text{ cm}^{-2}$.

We started our effort to model deposition with classical linear polymerization theory. Classical linear free radical polymerization kinetics (i.e., no branching or cross-linking) is well established in the solution phase,³⁸ where it has been demonstrated to apply to free radical polymerization of CHMA.^{36,37} While significant differences likely exist between iCVD polymerization and solution phase polymerization, existing studies of iCVD kinetics which propose complete quantitative models appeal to a model of the vapor phase which is incompatible with the results in Section 3.4 and used monomers which do not follow classical linear polymerization theory in solution^{5,8,9} (e.g., the monomer used is prone to branching^{39,40}). As such, these models are not suitable for our system. The kinetic equation for the rate of polymerization derived from classical linear free radical polymerization in solution can be adapted for iCVD by substituting surface concentrations instead of volumetric concentrations (similar assumptions are frequently made in kinetic models of iCVD^{5,21}) to give eq 22

$$r_p = k_p \sqrt{\frac{k_i}{k_t}} [I^*]^{1/2} [M]^{3/2} \quad (22)$$

where r_p is the rate of polymerization (deposition) in $\text{mol s}^{-1} \text{cm}^{-2}$, k_p is the propagation rate constant in $\text{cm}^2 \text{mol}^{-1} \text{s}^{-1}$, k_i is the initiation rate constant in $\text{cm}^2 \text{mol}^{-1} \text{s}^{-1}$, k_t is the termination rate constant in $\text{cm}^2 \text{mol}^{-1} \text{s}^{-1}$, $[I^*]$ is the surface concentration of initiating radicals in mol cm^{-2} , and $[M]$ is the surface concentration of monomer in mol cm^{-2} . Details of the derivation may be found in Odian.³⁸

While TBO* and Me* radicals are both present in the reactor, we assume that $[I^*]$ refers to TBO* radicals in eq 22. TBO* are frequently assumed to play a greater role in iCVD than Me* radicals due to their lower volatility.⁶ This idea is supported by data showing that an initiator with lower volatility than TBPO (*tert*-butyl peroxy-benzoate) has higher initiating efficiency than TBPO.¹⁴ Furthermore, a study of iCVD in nanoconfinement has found the initiator volatility ($p_i/p_{i,\text{sat}}$), necessary to describe kinetics.¹² In systems which are similar to iCVD, such as photoinitiated polymerization of butadiene vapor by TBPO, a strong preference for TBO* over Me* radicals has also been observed.^{30,53} While future work is needed to directly evaluate the factors that might drive preference for different initiators in iCVD, the existing work discussed provides reasonable support for our assumption.

To implement eq 22 in iCVD, relationships between the partial pressure and surface concentration are needed for CHMA and TBO*. While the BET isotherm is widely believed to describe monomer surface concentration in iCVD,^{3,17} and the Eley–Rideal model for initiator suggests a kinetic relationship like the Hertz–Knudsen equation,⁵⁴ both these equations can be approximated by a direct proportionality according to eq 23

$$[X] = k_X p_X \quad (23)$$

where $[X]$ is the surface concentration of species X in mol cm⁻² (CHMA or TBO*), p_X is the partial pressure in Pa and k_X (either k_{CHMA} or k_{TBPO}) is the proportionality-constant relating partial pressure to surface concentration in mol cm⁻² Pa⁻¹.

Though eq 21 provides a method of calculating the partial pressure of CHMA, we still need to establish a model for the partial pressure of initiating radicals. To obtain an approximate relationship for the partial pressure of TBO* radicals at the sample surface, we appeal to a mass balance on TBO*. As we established in Section 3.1, the concentration of TBO* radicals in the reactor is small, so the flow rate of TBO* radicals out of the reactor will be negligible compared to the rates of primary decomposition and β -scission. The mass balance on TBO* is therefore given by eq 24

$$r_1^{\text{tot}} = 2r_\beta^{\text{tot}} \quad (24)$$

where r_β^{tot} is the overall rate of β -scission occurring in the reactor in mol s⁻¹. To achieve our goal of estimating the partial pressure of TBO* radicals from eq 24, a relationship between r_β^{tot} and the partial pressure of TBO* radicals at the sample surface is needed. This comes in the form of a rate equation for β -scission analogous to eq 9, which we define based on eq 2 as

$$r_\beta^{\text{tot}} = k_\beta^{\text{eff}} \frac{p}{R_G T_{\text{fil}}} y_{\text{TBO}^*} \quad (25)$$

where k_β^{eff} is the effective rate constant for β -scission, defined analogously to k_1^{eff} and y_{TBO^*} is the mole fraction of TBO* radicals at the sample surface. When we substitute eq 9 for r_1^{tot} , and eq 25 for r_β^{tot} , we obtain eq 26

$$p_{\text{TBO}^*} = \frac{k_1^{\text{eff}}}{2k_\beta^{\text{eff}}} p_{\text{TBPO}} \quad (26)$$

Polymerization rate can now be obtained by substituting eqs 23 and 26 into eq 22 to obtain eq 27

$$r_p = k_{\text{surf}} p_{\text{TBPO}}^{1/2} p_{\text{CHMA}}^{3/2} \quad (27)$$

where $k_{\text{surf}} = k_p \sqrt{\frac{k_i k_{\text{TBO}^*}^3 k_{\text{CHMA}} k_1^{\text{eff}}}{2k_\beta^{\text{eff}}}}$ is a lumped rate constant for

deposition. Note that k_p , k_i , k_β , k_{TBO^*} , and k_{CHMA} all depend on stage temperature, while k_1^{eff} and k_β^{eff} depend on filament temperature, allowing us to consider these dependences independently. The work of Chan and Gleason¹⁷ investigates the stage temperature dependence of CHMA deposition with TBPO and their data suggest an approximate Arrhenius dependence over the temperature range of interest, such that the overall temperature dependence of k_{surf} can be approximated by eq 28

$$k_{\text{surf}} = A_{\text{surf}} \exp \left(-\frac{E_A^{\text{surf}}}{R_G T_{\text{stage}}} - \frac{E_{A,1}\eta_1 - E_{A,\beta}\eta_\beta}{2R_G T_{\text{fil}}} \right) \quad (28)$$

where A_{surf} is an empirical pre-exponential factor with units of mol cm⁻² s⁻¹ Pa⁻² and E_A^{surf} is the apparent activation energy describing the stage temperature dependence of deposition. Fitting the data of Chan and Gleason suggests an effective activation energy (E_A^{surf}) of 51 kJ mol⁻¹. Details of the fitting of data from Chan and Gleason may be found in the Supporting Information, and details on the mechanism underlying stage temperature dependence and a more precise model of stage temperature dependence are contained in Chan and Gleason.¹⁷

By rearranging eqs 27 and 28, the quantity $r_p p_{\text{TBPO}}^{-1/2} p_{\text{CHMA}}^{-3/2} \exp \left(\frac{E_A^{\text{surf}}}{R_G T_{\text{stage}}} \right)$ is expected to follow an Arrhenius relationship in the filament temperature. Figure 7 (c) shows a plot of $\frac{E_A^{\text{surf}}}{R_G T_{\text{stage}}} \ln(r_p p_{\text{TBPO}}^{-1/2} p_{\text{CHMA}}^{-3/2})$ versus $1000/T_{\text{fil}}$, showing a roughly linear trend ($r^2 = 0.77$). Deposition rate (r_p) was determined from FTIR absorption using the methods discussed above, p_{TBPO} was estimated using eq 20, and p_{CHMA} was estimated using eq 21. The slope of the linear fit shown in Figure 7 (c) corresponds to an apparent activation energy of 41.0 kJ mol⁻¹, and a pre-exponentiation factor $\ln(A_{\text{surf}})$ of -38.33 (mol Pa⁻² s⁻¹ cm⁻²). Comparison with the apparent activation energy for primary decomposition observed in Section 3.2 indicates that the apparent activation energy of β -scission is roughly 13.3 kJ mol⁻¹ ($\eta_\beta = 0.26$), significantly smaller than the literature value of 51 kJ mol⁻¹.⁵ To explain this, we consider that at uniform temperature β -scission is much less temperature-sensitive than primary decomposition. This means that in the reactor, it will not be as locally constrained to the filament array as primary decomposition, and the temperature of the filament will be a much poorer descriptor of the temperature at which β -scission is most frequently occurring (though still the closest we can measure). Further analysis of this effect would require theoretical or computational efforts to solve the simultaneous heat transfer, reaction, and diffusion problem at the array. The notable scatter in the data shown in Figure 7c is due to experimental error and suggests that this work should be taken as a preliminary indication of how the vapor phase modeling can be implemented in deposition models. Further work is needed, and we will likely find it necessary to modify the simple deposition model proposed here.

While the fit shown in Figure 7c is effective for relating the vapor phase model we propose to deposition over a range of

compositions and temperatures, it obscures how the proposed model relates to past work. Without the model of the vapor phase we develop, past efforts to investigate the dependence of iCVD on filament temperature have plotted deposition rate directly on Arrhenius axes. It was observed in many studies that at high filament temperatures ($>350\text{ }^{\circ}\text{C}$), the Arrhenius fit begins to fail, but no explanation was forthcoming.^{6,14} This type of plot is presented in Figure 8, where Figure 8a

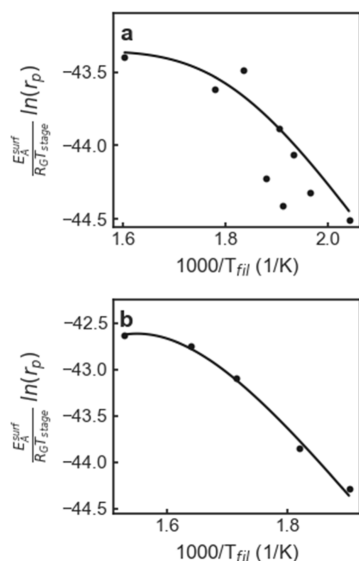


Figure 8. (a) Arrhenius plot of deposition rate data collected at $y_{\text{TBPO}}^{\text{in}}$ of 0.2 corrected for effects of changing stage temperature by data of Chan,¹⁷ plotted against the model fit of eq 27, which is also given as the dotted line in Figure 7c. (b) Arrhenius plot of deposition rate data from Xu and Gleason,¹⁴ similarly corrected for the effects of changing stage temperature and plotted against the model fit of eq 27.

reproduces our data for CHMA deposition with TBPO from Figure 7c at a $y_{\text{TBPO}}^{\text{in}}$ of 0.2, including the fit from eq 27. Note that the y-axis is identical to Figure 7c, except that the dependence on partial pressures of the monomer and initiator is removed. As a comparison with existing work, we have also reproduced the data of Xu and Gleason¹⁴ for the deposition of CHMA with TBPO at a range of filament temperatures, showing the characteristic kink at high temperatures.

Because the reactor system used by Xu and Gleason is not known to us and their process conditions differed somewhat from ours (both factors which impact TBPO decomposition), when fitting eq 27 to their data, η_1 , $\ln(A_1^{\text{eff}})$, η_β , and $\ln(A^{\text{surf}})$ were all varied using a gradient descent algorithm to minimize the MSE between the model and prediction. To obtain the solid line shown, fit values of 0.92, 22.8 ($\text{m}^3 \text{s}^{-1} \text{mol}^{-1}$), 0.27, and -17.9 ($\text{mol Pa}^{-2} \text{s}^{-1} \text{cm}^{-2}$) were obtained for η_1 , $\ln(A_1^{\text{eff}})$, η_β , and $\ln(A^{\text{surf}})$, respectively. The larger value of η_1 is likely due in part to the difference in temperature measurement, while the value for η_β matches well with our findings. Explaining the differences in pre-exponential factors requires further investigation.

Qualitatively, the kink in Figure 8 occurs at high temperatures because of the decomposition and mixing processes modeled in Sections 3.1–3.4. When the rate of TBPO decomposition is relatively small, it is primarily determined by the filament temperature, and since the rate of TBPO decomposition is proportional to the partial pressure of radicals (eq 26), deposition follows an Arrhenius dependence

under these conditions. However, when the rate of TBPO decomposition reaches the same order of magnitude as the total TBPO flow into the reactor, the rate of decomposition becomes limited by the amount of TBPO flowing into the reactor, and it begins to plateau. When the rate of TBPO decomposition plateaus, so does the deposition rate, causing the kink shown in Figure 8. Past work has hypothesized that the apparent change in activation energy occurs because, at low filament temperature the delivery of radicals to the surface is limited by the rate of decomposition, but at higher temperatures it is limited by mass transport to the surface (diffusion).⁶ The predictive success of our kinetic model based on an experimentally validated model of the vapor-phase suggests that diffusion is not the primary driver of this change in apparent activation energy. Furthermore, past descriptions of the effect have been largely qualitative,⁶ and our model is the first complete mathematical model which has been proposed to describe this phenomenon.

4. CONCLUSIONS

This study aims to provide a thorough characterization of the vapor phase during iCVD and relate this to the deposition rate. By elucidating the fundamental processes occurring in the vapor phase, we hope that our work can lay the groundwork for stronger models of the surface processes in iCVD.

Quantifying composition in the vapor phase begins with a thorough characterization of TBPO decomposition. During batch decomposition of TBPO, the change in reactor pressure can be used to determine the fractional conversion of TBPO. We use this to show that TBPO undergoes first-order decomposition with an Arrhenius temperature dependence in agreement with the literature on gas decomposition.^{28–30} This observation suggests that the rate of decomposition is limited by primary decomposition and that the radical species capable of initiating polymerization (TBO^* and Me^*) are only present in trace amounts. When we plot the mole fraction of TBPO against the Pirani gauge ratio, we observe a consistent linear relationship. When pure TBPO is decomposing, this relationship allows us to determine the real-time composition in the reactor from the Pirani gauge pressure.

The ability of the Pirani gauge to sense composition allows us to study the decomposition of TBPO as it flows through the reactor at steady-state, mirroring normal iCVD operation. Comparing composition determined by the Pirani gauge to the well-mixed assumption closely predicts decomposition kinetics over a range of inlet flow rates. Varying the temperature and pressure showed that decomposition follows Arrhenius kinetics, with very little pressure dependence above pressures of 50 mTorr. The pressure independence of kinetics is consistent with heat transfer experiments performed by Bakker et al.⁴⁵ in which it was found that heat transfer from the array in TBPO vapor varied little above 50 mTorr. While we have observed significant variability in temperature measurements, we estimate TBPO decomposition in our reactor to have an apparent activation energy of 91.5 kJ mol^{-1} ($\eta_1 = 0.578$) and $\ln(A_1^{\text{eff}})$ of 10.7 ($\text{m}^3 \text{s}^{-1}$). The difference between these kinetics and those at uniform temperature that is common in the literature arises because our kinetics (eq 9) are averaged over the nonuniform conditions of our reactor. Future efforts devoted to solving the heat and mass transfer problem close to the array are needed to accurately predict deposition from first principles.

Steady-state flow-through decomposition studies were also performed with mixed TBPO and CHMA at a range of inlet mole fractions. At a stage temperature of 50 °C, when there is no deposition, a model of the Pirani gauge ratio based only on the pure TBPO mixing, and unreacted CHMA/TBPO mixing is an excellent predictor of the Pirani gauge ratio. The excellent fit of the well-mixed model following first order Arrhenius kinetics to the measured Pirani gauge ratio provides no evidence to suggest that the monomer and initiator interact in the vapor phase; however, this possibility is not ruled out by our data. Since the monomer partial pressure is well established as a key parameter in iCVD kinetics, we compared the performance of our model accounting for thermal decomposition of TBPO, with the existing methods to show that significant error is incurred when these processes are ignored (up to 20% for the case plotted). When combined TBPO and CHMA decomposition experiments were performed at a stage temperature of 30 °C, slight overprediction of the Pirani gauge ratio was observed at conditions which favor deposition. This overprediction was used to estimate the overall rate of deposition in the reactor, and the effect of this on the monomer composition showed up to a 5% bias on the mole fraction of the monomer in the reactor. Such an effect could be important if a precise model is needed.

To relate the proposed model of the vapor phase to the deposition itself, depositions were performed at a range of filament temperatures and inlet mole fractions. Deposition rates were fitted to a simple model of linear (i.e., no branching) free radical polymerization in which TBO* was assumed to be the primary initiating species and stage temperature dependence was corrected using the data of Chan and Gleason.¹⁷ This model fit data reasonably well; however, there was significant experimental error. Based on this fitting, we were able to estimate the effective activation energy for β -scission in the reactor, which was 13.3 kJ mol⁻¹ ($\eta_\beta = 0.26$), suggesting that β -scission is much less localized to the array than primary decomposition. Importantly, our simple deposition model predicts a well-known decrease in the activation energy of iCVD deposition at high filament temperature (300 °C),^{6,14} illustrating its necessity in predicting deposition rate. To show this, we compare our deposition model to data from Xu and Gleason studying the same system,¹⁴ showing an excellent fit. Further work to describe transience in the vapor phase may be helpful for explaining the observed acceleration of deposition with pulsed filament iCVD,²⁶ as well as the transient behavior observed at the start of deposition.^{21,55}

While we have shown the remarkable utility of the Pirani gauge in quantifying vapor phase composition in an iCVD reactor, further work is needed to verify these results using alternative composition sensing techniques such as in situ FTIR or mass spectrometry. Specifically, work is needed to investigate the potential reactions between CHMA and TBPO in the vapor phase. Spectroscopic composition sensing methods may be able to detect reactions that were invisible to the Pirani gauge.

Though the use of thermocouples for measuring filament temperature is standard in iCVD, the issues with repeatability in this study (requiring recalibration of up to 100 °C) suggest a broader issue within the iCVD literature. In recognition of this, there is a need to develop noncontact temperature measurements based on the overall resistance of the filament or an infrared thermometer. Neither overall resistance nor radiation-based temperature measurements are established in iCVD and

will require a dedicated effort to develop and verify. A good implementation of a contact-free filament temperature measurement would allow for consistency in filament temperature measurements across research laboratories and reactors, significantly improving progress by ensuring published data can be compared.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.langmuir.5c01026>.

Details of the temperature calibration procedure, details of the effective temperature model, derivation of equations for batch decomposition, derivation of equations for continuous operation, derivation of equations for the Pirani gauge model for decomposing TBPO and CHMA, deposition conditions, additional explanation of MSE in variable angle spectroscopic ellipsometry data, and further discussion (PDF)

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Notes

The authors declare no competing financial interest.

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■ ABBREVIATIONS

FTIR, Fourier transform infrared
TBPO, di-*tert*-butyl peroxide
CHMA, cyclohexyl methacrylate
TBO*, *tert*-butoxy radical
Me*, methyl radical
iCVD, initiated chemical vapor deposition
CVD, chemical vapor deposition
BET, Brunauer Emmet Teller
ATR, attenuated total reflectance

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